

JAVA STRING PROGRAMS

Combined Practical Record

Program 1: Character Access – charAt()

Source Code

```
class StringOperation2 {  
    public static void main(String[] args) {  
        String str1 = "Hello";  
        String str2 = "Java";  
        String str3 = "World";  
  
        System.out.println(str1.charAt(0));  
        System.out.println(str2.charAt(1));  
        System.out.println(str3.charAt(2));  
    }  
}
```

1) EXECUTION STEPS

1. Start the program.
2. Create and initialize the required String variables.
3. Use charAt() with the specified index to access a character.
4. Print the returned characters.
5. Stop the program.

2) ALGORITHM

1. Start.
2. Initialize the String variables.
3. Call charAt(index) for the required positions.
4. Display the returned characters.
5. Stop.



TERMINAL — PROGRAM OUTPUT

```
$ javac StringOperation2.java  
$ java StringOperation2  
  
H  
a  
r
```

Program 3: String Comparison – compareTo()

Source Code

```
class Compare {  
  
    public static void main(String[] args) {  
        String str1 = "Apple";  
        String str2 = "Banana";  
        String str3 = "Apple";  
  
        System.out.println(str1.compareTo(str2));  
        System.out.println(str2.compareTo(str1));  
        System.out.println(str1.compareTo(str3));  
    }  
}
```

1) EXECUTION STEPS

1. Start the program.
2. Create and initialize the required String variables.
3. Compare the specified strings using compareTo().
4. Print the integer values returned by the comparisons.
5. Stop the program.

2) ALGORITHM

1. Start.
2. Initialize the strings.
3. Apply compareTo() to each specified pair.
4. Display the returned comparison values.
5. Stop.



TERMINAL — PROGRAM OUTPUT

```
$ javac Compare.java  
$ java Compare  
-1  
1  
0
```

Program 4: String Concatenation – concat()

Source Code

```
class Concat {  
    public static void main(String[] args) {  
        String str1 = "Hello";  
        String str2 = "Java";  
        String str3 = "World";  
  
        System.out.println(str1.concat(str2));  
        System.out.println(str2.concat(str3));  
        System.out.println(str1.concat(str3));  
    }  
}
```

1) EXECUTION STEPS	1. Start the program. 2. Create and initialize the required String variables. 3. Join the specified strings using concat(). 4. Print the resulting strings. 5. Stop the program.
2) ALGORITHM	1. Start. 2. Initialize the strings. 3. Apply concat() to the specified string pairs. 4. Display the concatenated strings. 5. Stop.



TERMINAL — PROGRAM OUTPUT

```
$ javac Concat.java  
$ java Concat  
  
HelloJava  
JavaWorld  
HelloWorld
```

Program 5: String Ending Check – endsWith()

Source Code

```
class End {  
  
    public static void main(String[] args) {  
        String str1 = "Hello World";  
        String str2 = "Java Programming";  
        String str3 = "Hello Java";  
  
        System.out.println(str1.endsWith("World"));  
        System.out.println(str2.endsWith("Java"));  
        System.out.println(str3.endsWith("Java"));  
    }  
}
```

1) EXECUTION STEPS	1. Start the program. 2. Create and initialize the required String variables. 3. Check whether each string ends with the specified suffix using endsWith(). 4. Print the Boolean results. 5. Stop the program.
2) ALGORITHM	1. Start. 2. Initialize the strings. 3. Apply endsWith() with the given suffixes. 4. Display true/false results. 5. Stop.

TERMINAL — PROGRAM OUTPUT

```
$ javac End.java  
$ java End  
  
true  
false  
true
```

Program 6: String Equality Check – equals()

Source Code

```
class Equal {  
  
    public static void main(String[] args) {  
        String str1 = "Java";  
        String str2 = "Java";  
        String str3 = "Python";  
  
        System.out.println(str1.equals(str2));  
        System.out.println(str1.equals(str3));  
        System.out.println(str2.equals(str3));  
    }  
}
```

1) EXECUTION STEPS

1. Start the program.
2. Create and initialize the required String variables.
3. Compare the strings using equals().
4. Print the Boolean results.
5. Stop the program.

2) ALGORITHM

1. Start.
2. Initialize the strings.
3. Apply equals() to the specified pairs.
4. Display true/false results.
5. Stop.

TERMINAL — PROGRAM OUTPUT

```
$ javac Equal.java  
$ java Equal  
  
true  
false  
false
```

Program 7: Case-Insensitive Equality – equalsIgnoreCase()

Source Code

```
class Ignore {  
  
    public static void main(String[] args) {  
        String str1 = "Java";  
        String str2 = "Hello";  
        String str3 = "java";  
  
        System.out.println(str1.equalsIgnoreCase(str2));  
        System.out.println(str1.equalsIgnoreCase(str3));  
        System.out.println(str2.equalsIgnoreCase(str3));  
    }  
}
```

1) EXECUTION STEPS	1. Start the program. 2. Create and initialize the required String variables. 3. Compare the strings without considering uppercase/lowercase differences using equalsIgnoreCase(). 4. Print the Boolean results. 5. Stop the program.
2) ALGORITHM	1. Start. 2. Initialize the strings. 3. Apply equalsIgnoreCase() to the specified pairs. 4. Display true/false results. 5. Stop.

TERMINAL — PROGRAM OUTPUT

```
$ javac Ignore.java  
$ java Ignore  
false  
true  
false
```

Program 8: String Search – indexOf()

Source Code

```
class Index {  
  
    public static void main(String[] args) {  
        String str1 = "Hello";  
        String str2 = "Java";  
        String str3 = "Programming";  
  
        System.out.println(str1.indexOf('l'));  
        System.out.println(str2.indexOf('v'));  
        System.out.println(str3.indexOf("gram"));  
    }  
}
```

1) EXECUTION STEPS

1. Start the program.
2. Create and initialize the required String variables.
3. Search for the specified character or substring using indexOf().
4. Print the returned index values.
5. Stop the program.

2) ALGORITHM

1. Start.
2. Initialize the strings.
3. Apply indexOf() to the specified characters/substrings.
4. Display the returned positions.
5. Stop.

TERMINAL — PROGRAM OUTPUT

```
$ javac Index.java  
$ java Index  
2  
3  
3
```

Program 9: String Length – length()

Source Code

```
class Length {  
  
    public static void main(String[] args) {  
        String str1 = "Hello";  
        String str2 = "Java";  
        String str3 = "Programming";  
  
        System.out.println("Length of str1: " + str1.length());  
        System.out.println("Length of str2: " + str2.length());  
        System.out.println("Length of str3: " + str3.length());  
    }  
}
```

1) EXECUTION STEPS

1. Start the program.
2. Create and initialize the required String variables.
3. Find the number of characters in each string using length().
4. Print the lengths.
5. Stop the program.

2) ALGORITHM

1. Start.
2. Initialize the strings.
3. Apply length() to each string.
4. Display the number of characters.
5. Stop.



TERMINAL — PROGRAM OUTPUT

```
$ javac Length.java  
$ java Length  
  
Length of str1: 5  
Length of str2: 4  
Length of str3: 11
```


Program 10: Character Replacement – replace()

Source Code

```
class Replace {  
  
    public static void main(String[] args) {  
        String str1 = "Hello";  
        String str2 = "Java";  
        String str3 = "World";  
  
        System.out.println(str1.replace('l', 'x'));  
        System.out.println(str2.replace('a', 'o'));  
        System.out.println(str3.replace('o', 'a'));  
    }  
}
```

1) EXECUTION STEPS

1. Start the program.
2. Create and initialize the required String variables.
3. Replace the specified character with the new character using replace().
4. Print the modified strings.
5. Stop the program.

2) ALGORITHM

1. Start.
2. Initialize the strings.
3. Apply replace(old, new) to each specified string.
4. Display the modified strings.
5. Stop.



TERMINAL — PROGRAM OUTPUT

```
$ javac Replace.java  
$ java Replace  
  
Hexxo  
Jovo  
Warld
```

Program 11: String Beginning Check – startsWith()

Source Code

```
class Starting {  
  
    public static void main(String[] args) {  
        String str1 = "Hello World";  
        String str2 = "Java Programming";  
        String str3 = "Hello Java";  
  
        System.out.println(str1.startsWith("Hello"));  
        System.out.println(str2.startsWith("Java"));  
        System.out.println(str3.startsWith("World"));  
    }  
}
```

1) EXECUTION STEPS	1. Start the program. 2. Create and initialize the required String variables. 3. Check whether each string starts with the specified prefix using startsWith(). 4. Print the Boolean results. 5. Stop the program.
2) ALGORITHM	1. Start. 2. Initialize the strings. 3. Apply startsWith() with the given prefixes. 4. Display true/false results. 5. Stop.

TERMINAL — PROGRAM OUTPUT

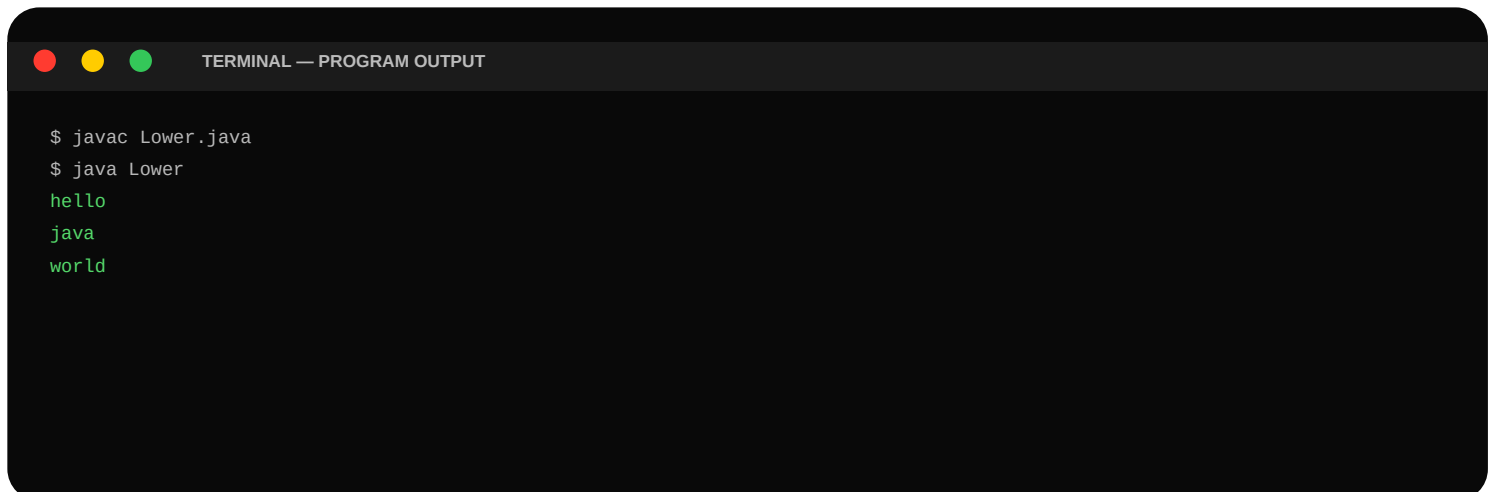
```
$ javac Starting.java  
$ java Starting  
  
true  
true  
false
```

Program 12: String Conversion to Lowercase – toLowerCase()

Source Code

```
class Lower {  
  
    public static void main(String[] args) {  
        String str1 = "HELLO";  
        String str2 = "JAVA";  
        String str3 = "WORLD";  
  
        System.out.println(str1.toLowerCase());  
        System.out.println(str2.toLowerCase());  
        System.out.println(str3.toLowerCase());  
    }  
}
```

1) EXECUTION STEPS	1. Start the program. 2. Create and initialize the required String variables. 3. Convert each string into lowercase using toLowerCase(). 4. Print the converted strings. 5. Stop the program.
2) ALGORITHM	1. Start. 2. Initialize the String variables. 3. Call toLowerCase() for each string. 4. Display the converted strings. 5. Stop.

A terminal window titled "TERMINAL — PROGRAM OUTPUT" with a dark background. It shows the compilation and execution of the Java program. The commands entered are \$ javac Lower.java and \$ java Lower. The output displayed is hello, java, and world on separate lines.

```
TERMINAL — PROGRAM OUTPUT  
  
$ javac Lower.java  
$ java Lower  
hello  
java  
world
```

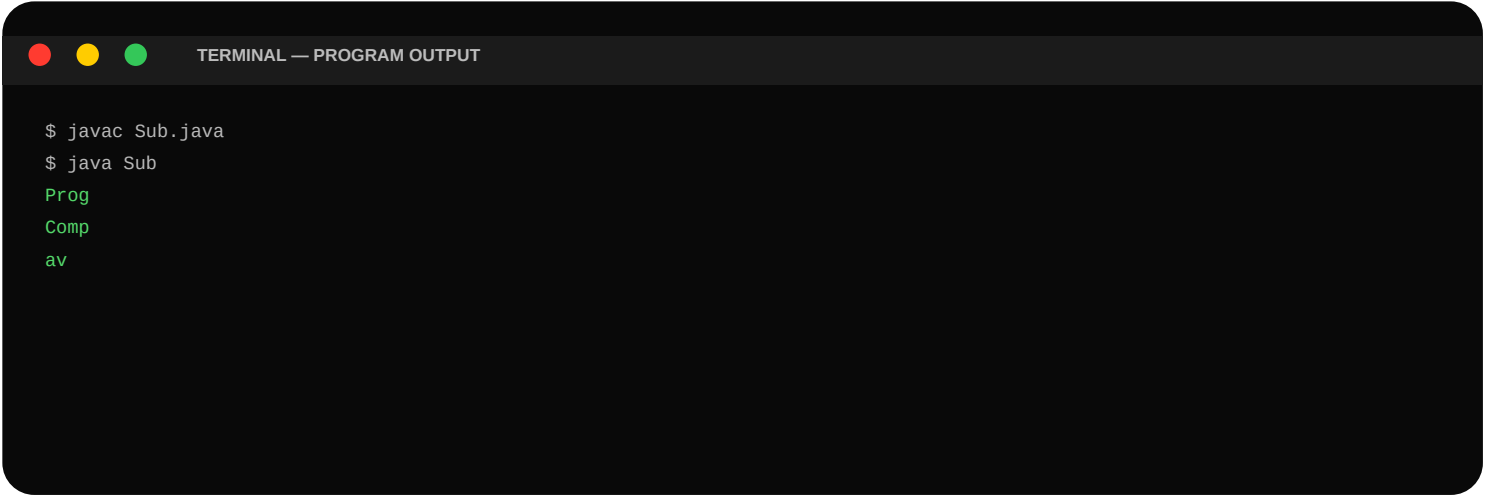
Program 13: String Extraction – substring()

Source Code

```
class Sub {
    public static void main(String[] args) {
        String str1 = "Programming";
        String str2 = "Computer";
        String str3 = "Java";

        System.out.println(str1.substring(0,4));
        System.out.println(str2.substring(0,4));
        System.out.println(str3.substring(1,3));
    }
}
```

1) EXECUTION STEPS	1. Start the program. 2. Create and initialize the required String variables. 3. Extract the required parts using substring(). 4. Print the extracted strings. 5. Stop the program.
2) ALGORITHM	1. Start. 2. Initialize the strings. 3. Apply substring() to the specified strings. 4. Display the extracted strings. 5. Stop.

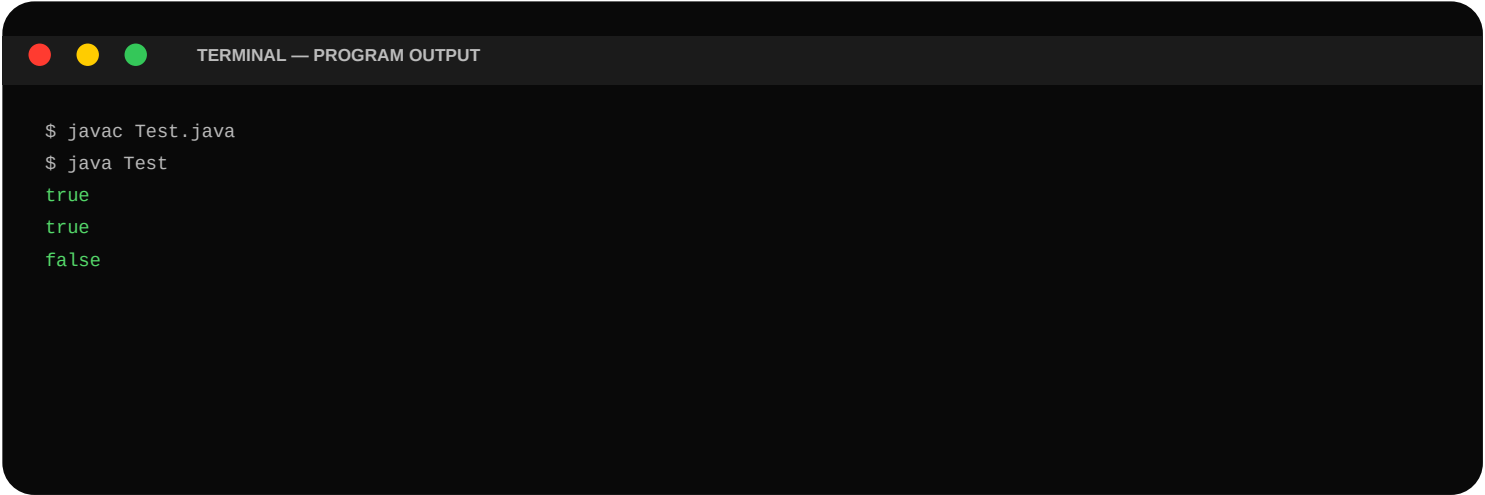


Program 14: String Search – contains()

Source Code

```
class Test {  
  
    public static void main(String[] args) {  
        String str1 = "Hello World";  
        String str2 = "Java Programming";  
        String str3 = "Hello Java";  
  
        System.out.println(str1.contains("World"));  
        System.out.println(str2.contains("Java"));  
        System.out.println(str3.contains("Python"));  
    }  
}
```

1) EXECUTION STEPS	1. Start the program. 2. Create and initialize the required String variables. 3. Search for the specified substring using contains(). 4. Print the Boolean results. 5. Stop the program.
2) ALGORITHM	1. Start. 2. Initialize the strings. 3. Apply contains() to the specified strings. 4. Display the true/false results. 5. Stop.

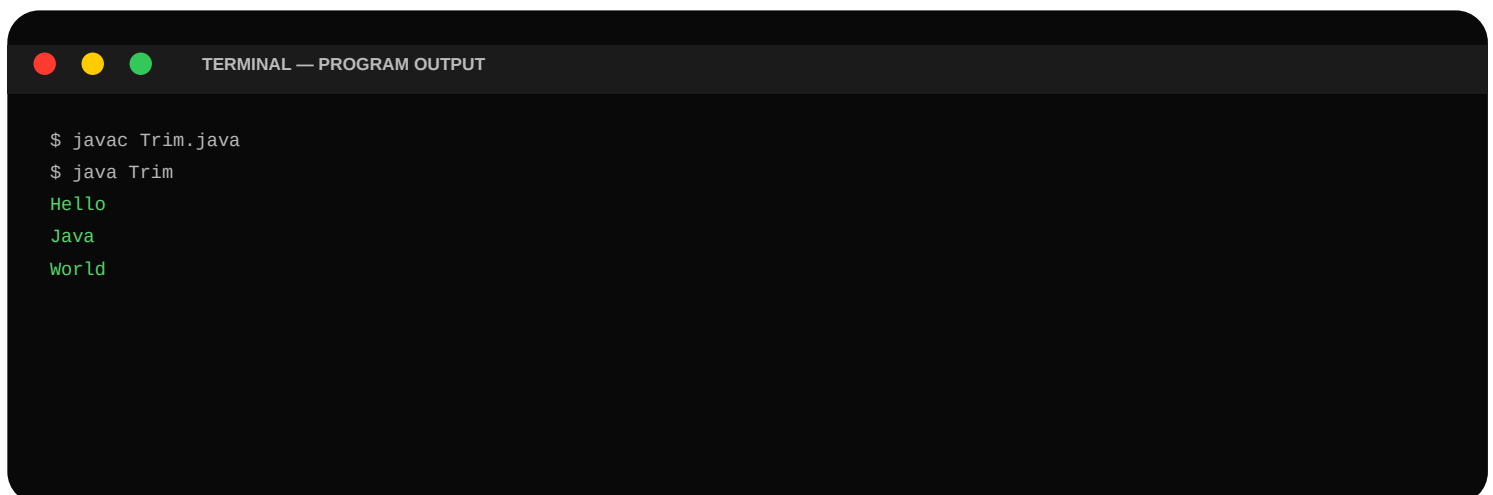


Program 15: String Trimming – trim()

Source Code

```
class Trim {  
  
    public static void main(String[] args) {  
        String str1 = " Hello ";  
        String str2 = " Java ";  
        String str3 = " World ";  
  
        System.out.println(str1.trim());  
        System.out.println(str2.trim());  
        System.out.println(str3.trim());  
    }  
}
```

1) EXECUTION STEPS	1. Start the program. 2. Create and initialize the required String variables. 3. Remove leading and trailing spaces using trim(). 4. Print the trimmed strings. 5. Stop the program.
2) ALGORITHM	1. Start. 2. Initialize the strings. 3. Apply trim() to each string. 4. Display the trimmed strings. 5. Stop.

A terminal window titled "TERMINAL — PROGRAM OUTPUT" with three colored window control buttons (red, yellow, green) on the left. The terminal shows the execution of a Java program. The first two lines are commands: "\$ javac Trim.java" and "\$ java Trim". The next three lines are the program's output: "Hello", "Java", and "World", each on a new line.

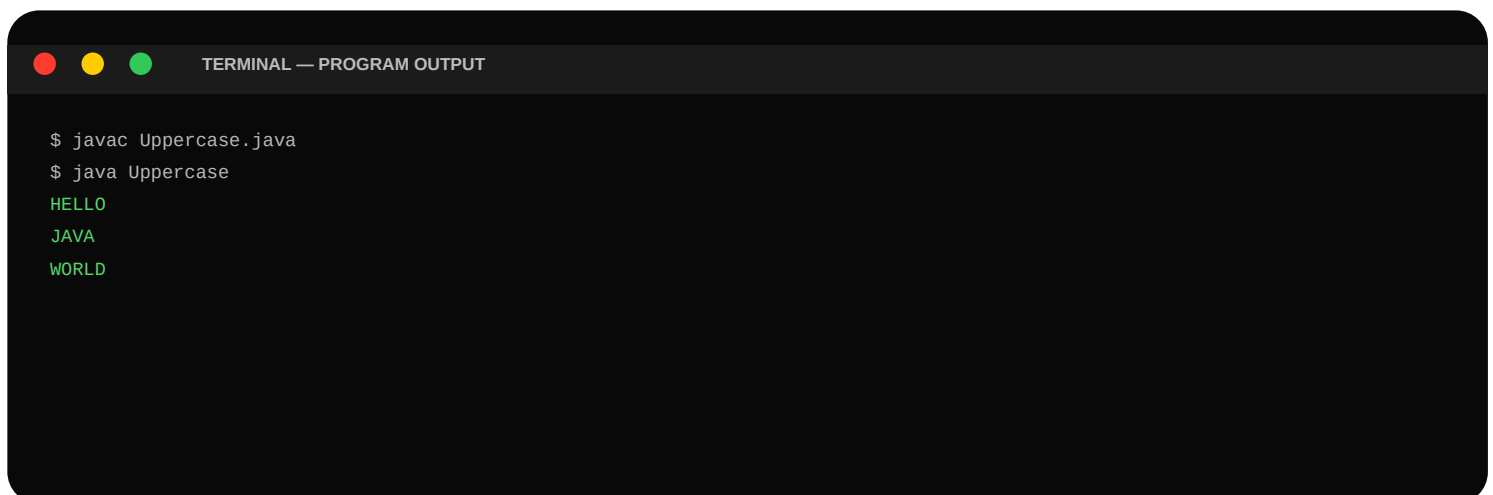
```
TERMINAL — PROGRAM OUTPUT  
  
$ javac Trim.java  
$ java Trim  
Hello  
Java  
World
```

Program 16: String Conversion to Uppercase – toUpperCase()

Source Code

```
class Uppercase {  
  
    public static void main(String[] args) {  
        String str1 = "hello";  
        String str2 = "java";  
        String str3 = "world";  
  
        System.out.println(str1.toUpperCase());  
        System.out.println(str2.toUpperCase());  
        System.out.println(str3.toUpperCase());  
    }  
}
```

1) EXECUTION STEPS	1. Start the program. 2. Create and initialize the required String variables. 3. Convert each string into uppercase using toUpperCase(). 4. Print the converted strings. 5. Stop the program.
2) ALGORITHM	1. Start. 2. Initialize the String variables. 3. Call toUpperCase() for each string. 4. Display the converted strings. 5. Stop.

A terminal window titled "TERMINAL — PROGRAM OUTPUT" with three colored window control buttons (red, yellow, green) on the left. The terminal shows the compilation and execution of the Uppercase.java program. The output displays the words "HELLO", "JAVA", and "WORLD" in green text on separate lines.

```
$ javac Uppercase.java  
$ java Uppercase  
HELLO  
JAVA  
WORLD
```